Space for rough work

(a) (i) → (D); (ii) → (B); (iii) → (A); (iv) → (C)
(b) (i) → (B); (ii) → (D); (iii) → (A); (iv) → (C)
(c) (i) → (D); (ii) → (C); (iii) → (B); (iv) → (A)
(d) (i) → (B); (ii) → (C); (iii) → (D); (iv) → (A)

6. The possible chain isomers of heptyne is/are__________.
(a) 1 (b) 3 (c) 4 (d) 2

7. Identify the pair of position isomers of 2-methyl 1-pentene.
(a) 3-methyl 1-pentene and 2-methyl 2-pentene
(b) 3-methyl 1-pentene and 3-methyl 2-pentene
(c) 4-methyl 1-pentene and 4-methyl 2-pentene
(d) All of the above

8. Identify the aliphatic hydrocarbon with one triple bond.
(a) C_3H_6 (b) C_4H_8 (c) C_5H_8 (d) C_2H_6

9. The vapour density of an aliphatic unsaturated hydrocarbon 'X' is 13. Determine the number of bromine atoms required to make it saturated and identify X.
(a) 2, X → Ethane (b) 4, X → Ethene
(c) 2, X → Ethyne (d) 4, X → Ethyne

10. Identify the true statement among the following.
(a) 1-butyne and 1-pentene are homologues
(b) 1-butyne and 2-butyne are homologues
(c) 1-butene and 2-butyne are homologues
(d) 1-pentyne and 2-hexyne are homologues

11. Excess of methane, on treatment with chlorine in presence of sunlight, gives _________ and _________.
(a) CCl_4 and HCl (b) CH_3Cl and HCl
(c) $CHCl_3$ and HCl (d) CH_2Cl_2 and HCl

12. Sodium salt of acetic acid on treatment with soda lime gives hydrocarbon 'X'. X can also be prepared by passing a mixture of 'Y' and 'Z' over heated nickel carbonate at 250°C Identify X, Y and Z.
(a) $X \rightarrow CH_4$, $Y \rightarrow CO_2$, $Z \rightarrow H_2$
(b) $X \rightarrow CH_4$, $Y \rightarrow CO$, $Z \rightarrow H_2O$
(c) $X \rightarrow CH_4$, $Y \rightarrow CO_2$, $Z \rightarrow H_2O$
(d) $X \rightarrow C_2H_4$, $Y \rightarrow CO$, $Z \rightarrow H_2O$

13. Study the following reaction and identify the colour of 'A' and 'B'.

$$HC \equiv CH \xrightarrow{AgNO_3 + NH_4OH} \text{Precipitate A}$$
$$HC \equiv CH \xrightarrow{CuCl} \text{Precipitate B}$$

Ethyne

(a) A → White, B → Black
(b) A → Black, B → White
(c) A → White, B → Red
(d) A → Red, B → White

14. During the fractional distribution of coal tar, the temperature range in which phenol is obtained is _________.

(a) 80°C–230°C (b) 170°C–270°C
(c) 230°C–400°C (d) 270°C–400°C

15. The molar ratio of CO_2 and H_2O in the complete combustion of hydrocarbon 'X' is 10:8. Calculate the percentage of carbon in X.

(a) 86.4% (b) 88.2% (c) 85.7% (d) 83.3%

Space for rough work

Answer Keys

Assessment Test I

1. (d) **2.** (a) **3.** (c) **4.** (d) **5.** (a) **6.** (a) **7.** (d) **8.** (b) **9.** (a) **10.** (d)
11. (c) **12.** (a) **13.** (a) **14.** (d) **15.** (d)

Assessment Test II

1. (c) **2.** (a) **3.** (b) **4.** (c) **5.** (c) **6.** (c) **7.** (d) **8.** (a) **9.** (c) **10.** (b)
11. (c) **12.** (c) **13.** (a) **14.** (c) **15.** (a)

Assessment Test III

1. (b) **2.** (a) **3.** (b) **4.** (d) **5.** (b) **6.** (d) **7.** (d) **8.** (b) **9.** (c) **10.** (a)
11. (a) **12.** (c) **13.** (a) **14.** (d) **15.** (b)

Assessment Test IV

1. (d) **2.** (c) **3.** (a) **4.** (a) **5.** (c) **6.** (c) **7.** (d) **8.** (b) **9.** (c) **10.** (b)
11. (b) **12.** (b) **13.** (d) **14.** (d) **15.** (d)

Hints and Explanations

CHAPTER 1
Nature of Matter

Assessment Test I

1. The water in the test tube boils if it contains distilled water kept in a container with boiling sea water. This happens because sea water has higher boiling point. Hence, the temperature of distilled water can reach 100°C and still the transmission of heat is possible because of difference in temperature.

   Hence, the correct option is (a).

2. Cooling of water in an earthen pot, usage of perfume and usage of air cooler are based on the principle that evaporation produces cooling. However, fogging on spectacles after coming out of an air-conditioned car is due to condensation of water vapour on the surface of glass which remains cool due to the air-conditioner.

   Hence, the correct option is (c).

3. Regelation is the process involved in passing a metallic wire through ice block.

   Hence, the correct option is (a).

4. Water exists in liquid state without freezing on mountain peaks even when the temperature falls slightly below the freezing point. This is because the freezing point of water decreases to less than 0°C due to the decrease in external pressure at high altitudes.

   Hence, the correct option is (c).

5. (i) Fractional distillation
   (ii) Fractional crystallization
   (iii) Solvent extraction
   (iv) Centrifugation

   Hence, the correct option is (c).

6. (i) Distilled water at hill station
   (ii) Distilled water at sea level
   (iii) Drinking water at sea level
   (iv) Sea water at sea level

   Hence, the correct option is (b).

7. Separation of components of blood is carried out by centrifugation whereas all others are applications of chromatography.

   Hence, the correct option is (a).

8. Ethyl alcohol and water differ with reference to boiling pints and they are miscible liquids. Therefore, they can be separated by fractional distillation and not by separating funnel.

   Hence, the correct option is (d).

9. Sulphur + sand mixture cannot be separated by the method of sublimation whereas the other mixtures can be separated by the method of sublimation.

   Hence, the correct option is (c).

10. The inferences which can be drawn on the basis of the above facts are that X and Y are immiscible liquids and X and water can be separated by fractional distillation.

    Hence, the correct option is (c).

11. The amount of component that can be separated by cooling up to any temperature depends upon the solubility of that substance at that temperature.

Hence, the correct option is (c).

12. N_2 and O_2 can be separated by fractional evaporation SO_2 and N_2 can be separated by dissolution in KOH. H_2 and NH_3 can be separated by preferential liquefaction. H_2 and Ar can be separated by diffusion.NH_3 and O_2 can be separated by dissolution in water.

(a) $\rightarrow$ (iv); (b) $\rightarrow$ (v); (c) $\rightarrow$ (i); (d) $\rightarrow$ (ii); (e) $\rightarrow$ (iii)

Hence, the correct option is (c).

13. $NaNO_3$ and NaCl can be separated by fractional crystallization. This is because the solubility of KNO_3 decreases to a large extent on cooling. However, the decrease in solubility of NaCl with the decrease in temperature is insignificant.

Hence, the correct option is (b).

14. A mixture of SO_2 and water cannot be separated by heating as it forms sulphurous acid. SO_2 is highly soluble in water.

Hence, the correct option is (d).

15. Refining of petroleum involves the separation of various components by fractional distillation as they possess different boiling points.

Hence, the correct option is (b).

Assessment Test II

1. Sea water has lower freezing point than normal water. Since ice is at the temperature of 0°C, further transfer of heat from sea water to ice is not possible and hence, phase transition is not possible. Distilled water in ice also does not undergo phase transition as there is no more heat transfer after reaching 0°C.

Hence, the correct option is (a).

2. Evaporation is a surface phenomenon which leads to decrease in temperature of surroundings.

Hence, the correct option is (c).

3. Skating on ice is based on the principle of regelation, whereas all other processes are based on condensation.

Hence, the correct option is (a).

4. Due to the presence of dissolved salts, the freezing point of water decreases and hence, water freezes at below 0°C.

Hence, the correct option is (c).

5. (i) Sand + chalk powder

(ii) Blood

(iii) Water + alcohol

(iv) KNO_3 + water

(v) Petrol + alcohol

Hence, the correct option is (a).

6. (i) Cooking in open vessel at hill station

(ii) Cooking in a vessel covered with lid at hill station

(iii) Cooking in pressure cooker at hill station

(iv) Cooking in pressure cooker at sea level

Hence, the correct option is (d).

7. Separation of leaf pigments can be done by the method of chromatography. The terms stationary phase, mobile phase and adsorpotion are related to chromatography whereas the term distillate is related to distillation.

Hence, the correct option is (a).

8. In the process of fractional distillation, the liquid with lower boiling point (X) is called distillate and is collected in the receiver flask.

Hence, the correct option is (a).

9. The figure represents the process of sublimation of a mixture with one sublimely component and another non-sublimely component. Among the given mixtures, a mixture of ammonium chloride and sand can be separated by this method.

Hence, the correct option is (d).

10. (A) and (D) are true statements. The separation method is separation of immiscible liquids. A

and B are, therefore, immiscible liquids and the denser liquid B forms the bottom layer.

Hence, the correct option is (c).

11. On cooling the hot-saturated solution of X and Y, Y being more soluble at 30°C than X at the same temperature, crystallizes first leaving behind the other component.

Hence, the correct option is (d).

12. Preferential liquefaction is based on the difference in critical temperatures. Fractional evaporation is based on the difference in boiling points of liquids of the corresponding gases. Diffusion is based on the difference in the molecular masses of gases. Dissolution in a suitable solvent is based on preferential solubility of a gas in a solvent.

(a) → (iii); (b) → (iv); (c) → (i); (d) → (ii);

Hence, the correct option is (a).

13. NaCl and C can be separated by dissolution in water as NaCl is highly soluble in water and C is insoluble in water.

Hence, the correct option is (c).

14. On opening a Pepsi bottle, CO_2 comes out with a fizzing sound as the solubility of gas decreases with the decrease in pressure.

Hence, the correct option is (c).

15. Making of coffee decoction involves the principle of solvent extraction. Refining of petrol and purification of ethyl alcohol are based on fractional distillation.

Hence, the correct option is (b).

Assessment Test III

1. Rate of evaporation is directly proportional to wind speed and inversely proportional to humidity.

Hence, the correct option is (c).

2. In a compound, only one type of molecules is present and it is uniformly distributed.

Hence, the correct option is (a).

3. (i) → (D); (ii) → (A); (iii) → (E); (iv) → (F)

Hence, the correct option is (a).

4. The liquid mixture is taken in a separating funnel.

The mixture is kept still.

Clear layers of the liquids are formed and the heavier liquid, that is, water forms the bottom layer.

The heavier component is collected first by opening the stop lock.

Hence, the correct option is (a).

5. In solids, molecules possess a very low kinetic energy.

Hence, the correct option is (d).

6. On cooling, the gas molecules lose heat energy which decreases the kinetic energy as well as potential energy of the molecules.

Further extraction of heat results in the decrease in potential energy of molecules whereas kinetic energy remains constant.

Decrease in potential energy leads to increase in intermolecular forces of attraction.

Increase in forces of attraction results in the conversion of state.

Hence, the correct option is (b).

7. Addition of impurities to the solids decreases the melting point of solids.

Hence, the correct option is (c).

8. Gas molecules are very loosely packed. Hence, they are bad conductors of heat.

Hence, the correct option is (a).

9. The appearance of fog is due to the condensation of water vapour into tiny droplets due to very low temperatures in winter morning.

Hence, the correct option is (c).

10. In a mixture of iron and sand, iron is magnetic and sand is non-magnetic in nature. Therefore, it can be separated by magnetic separation method.

Hence, the correct option is (b).

11. Water rises in a capillary tube due to strong adhesive forces between glass molecules and water molecules.

    Hence, the correct option is (a).

12. Since CO_2 is acidic in nature it reacts with KOH.

    Hence, the correct option is (c).

13. Critical temperature

    $A \rightarrow 32°C$

    $B \rightarrow -118°C$

    $C \rightarrow 31.1°C$

    $D \rightarrow -140°C$

    We know, critical temperature is directly proportional to intermolecular force which is directly proportional to ease of liquefaction.

    So, the order will be

    $D < B < C < A$

    Hence, the correct option is (a).

14. The melting points of silver and gold, which expand on melting, increase with the increase in pressure on them.

    Hence, the correct option is (b).

15. Naphthalene undergoes sublimation at room temperature and hence, it disappears with time.

    Hence, the correct option is (b).

Assessment Test IV

1. Water present in the wet cloth undergoes evaporation which causes cooling.

   Hence, the correct option is (a).

2. Pure substances are made up of only one kind of molecules.

   Hence, the correct option is (a).

3. (i) → (B); (ii) → (C); (iii) → (E); (iv) → (F)

   Hence, the correct option is (c).

4. The required order is CADB.

   Hence, the correct option is (b).

5. Copper exists in solid form and in solids molecules possess high intermolecular forces of attraction.

   Hence, the correct option is (d).

6. The required order is BCA.

   Hence, the correct option is (c).

7. Pressure cooker reduces cooking time because pressure inside the cooker is more, so boiling point of water increases. The temperature of the cooking medium being greater than the normal boiling point, food gets cooked at a faster rate, thus saving time and fuel.

   Hence, the correct option is (c).

8. Gas molecules show maximum fluidity because they have high kinetic energy and very less intermolecular forces of attraction.

   Hence, the correct option is (b).

9. Rate of evaporation is proportional to the surface area of the liquid.

   Hence, the correct option is (c).

10. In the mixture of NaCl and NH_4Cl, NH_4Cl sublimes on heating. Therefore, this mixture can be separated by the process of sublimation.

    Hence, the correct option is (c).

11. Cotton clothes are preferred in summer because they absorb sweat more and allow the sweat to be evaporated at a faster rate. The sweat absorbs heat energy from the body and gets evaporated, thus keeping the body cool.

    Hence, the correct option is (b).

12. Since N_2 and O_2 differ in boiling points in their liquid state, they can be separated by fractional evaporation.

    Hence, the correct option is (d).

13. Critical temperature is the temperature above which a gas cannot be liquified, no matter what amount of high pressure is applied on it.

    Hence, the correct option is (d).

14. Regelation involves the effect of pressure on melting point of ice.

    Hence, the correct option is (d).

15. Solid carbon dioxide undergoes sublimation due to high kinetic energy of surface molecules.

    Hence, the correct option is (c).

CHAPTER 2
Atomic Structure

Assessment Test I

1. The correct electronic configuration is 2, 8, 18 and 1.

   Hence, the correct option is (a).

2. The given configuration is a, $4a$, $(6a + 1)$, and $(a - 1)$.

   Since $a = 2$, the electronic configuration is 2, 8, 13, and 1.

   This is the electronic configuration of chromium.

   Hence, the correct option is (b).

3. Discovery of radioactivity proves the existence of subatomic particles. Therefore, it can be concluded that atoms are divisible.

   Hence, the correct option is (b).

4. Given that $Z_x{:}Z_y = 1{:}7$

   $r_3 = \dfrac{0.529\times n^2}{Z_x}$

   $\Rightarrow 1.587 = \dfrac{0.529\times 9}{Z_x}$

   $\Rightarrow Z_x = 3$

   $\therefore$ Atomic number of Y is 21.

   $\therefore$ Electronic configuration of Y is 2, 8, 9 and 2.

   Hence, the correct option is (b).

5. Electronic configuration of X is 2, 8, 18 and 8.

   Electronic configuration of A is 2, 8, 18, 8 and 2.

   $\therefore$ No. of electrons in the antepenultimate shell = 18

   Hence, the correct option is (a).

6. According to Rutherford's atomic model, electrons revolve around the nucleus with high velocities.

   Hence, the correct option is (d).

7. In α-ray scattering experiment, α-rays are used because they are positively charged particles and have mass.

   Hence, the correct option is (d).

8. $\dfrac{3A+2(A+x)}{5}=A+0.8$

   $\therefore 5A + 2x = 5A + 4$

   $\therefore 2x = 4 \qquad \therefore x = 2$

   $\therefore$ The mass of heavier isotope is $A + 2$.

   Hence, the correct option is (b).

9. $-13.6=\dfrac{-13.6\times Z^2}{9}$

   $\therefore Z = 3$

   $\therefore r_3 = \dfrac{0.529\times 3^2}{Z} = \dfrac{0.529\times 3^2}{3} = 0.529\times 3$

   =1.587 A

   Hence, the correct option is (a).

10. $E_2 = \dfrac{-13.6\times Z^2}{n^2}$

    $\therefore -x = \dfrac{-13.6\times Z^2}{4}$

    $\therefore Z^2 = \dfrac{4x}{13.6}$

    $\therefore Z = 2\sqrt{\dfrac{x}{13.6}}$

    Hence, the correct option is (b).

11. Atomic number of A is 49.

    Atomic number of B is 51.

    Atomic number of C is 53.

    Atomic number of D is 48.

    $\therefore$ The order is DABC.

    Hence, the correct option is (c).

12. Valence electron of A is 2.

    Valence electron of B is 6.

    Valence electron of C is 5.

    Valence electron of D is 1.

    $\therefore$ The order is DACB.

    Hence, the correct option is (b).

13. 8 electrons in the last orbit make the atom stable (unreactive) because they remain in low-energy state.

Hence, the correct option is (c).

14. From J. J. Thomson's experiment, $\frac{e}{m}$ ratio (specific charge) can be calculated and from Millikan's oil drop experiment, charge of an electron can be calculated. Hence, dividing e by $\frac{e}{m}$, mass of the electron can be calculated.

Hence, the correct option is (a).

15. 5th orbit to 3rd orbit $\rightarrow$ 3

4th orbit to 1st orbit $\rightarrow$ 6

6th orbit to 2nd orbit $\rightarrow$ 10

5th orbit to 4th orbit $\rightarrow$1

(i) $\rightarrow$ (D), (ii) $\rightarrow$ (C), (iii) $\rightarrow$ (A), (iv) $\rightarrow$ (B)

Hence, the correct option is (d).

Assessment Test II

1. The electronic configuration of the element is 2, 8, 14, and 2.

$\therefore$ The element is iron.

Hence, the correct option is (a).

2. The electronic configuration of zinc is 2, 8, 18, 2 [x, ($x + y$), 3y, x].

Hence, the correct option is (a).

3. Atoms of same element are identical in all aspects.

Hence, the correct option is (b).

4. Given that $Z_p : Z_Q = 2:7$

$$r_4 = \frac{-13.6 \times Z^2}{n^2}$$

$$\Rightarrow -13.6 = \frac{-13.6 \times Z_p{}^2}{4^2} \Rightarrow Z_p = 4$$

$\therefore Z_Q = 14$

Electronic configuration of Q is 2, 8, and 4.

$\therefore$ Difference in number of electrons = 8 – 4 = 4

Hence, the correct option is (b).

5. Electronic configuration of X is 2, 8 and 5.

$\therefore$ Electronic configuration of Y is 2, 8 and 8.

Hence, the correct option is (b).

6. According to Bohr's atomic model, a fixed amount of energy is emitted when an electron jumps from an outer orbit to an inner orbit.

Hence, the correct option is (d).

7. α-ray scattering experiment was carried out to prove the correctness of J. J. Thomson's atomic model.

Hence, the correct option is (c).

8. $$\frac{A \times m + (A+2)n}{m+n} = A + 1.25$$

where A is atomic weight of lighter isotope.

$(A + Z)$ is atomic weight of heavier isotope

m is % abundance of lighter isotope.

n is % abundance of heaver isotope.

$\therefore mA + nA + 2n = mA + nA + 1.25m + 1.25n$

$\therefore 0.75n = 1.25m$

$$\therefore \frac{m}{n} = \frac{0.75}{1.25} = \frac{3}{5}$$

Hence, the correct option is (c).

9. $$r_3 = \frac{0.529 \times n^2}{Z}$$

$$\therefore x = \frac{0.529 \times 9}{Z}$$

$$\therefore Z = \frac{4.761}{x}$$

Hence, the correct option is (a).

10. $$r_4 = \frac{0.529 \times 4^2}{Z} \mathring{A}$$

$$\therefore 2.116 = \frac{0.529 \times 4^2}{Z}$$

$$\therefore Z = \frac{0.529 \times 4^2}{2.116} = 4$$

$$\therefore E_4 = \frac{-13.6 \times n^2}{Z^2} \text{eV} = \frac{-13.6 \times 4^2}{4^2} = -13.6 \text{ eV}$$

Hence, the correct option is (b).

11. Atomic number of P is 24.

Atomic number of Q is 32.

Atomic number of R is 19.

Atomic number of S is 36.

The order is DBAC.

Hence, the correct option is (d).

12. Number of electrons in the outermost shell of P is 2.

Number of electrons in the outermost shell of Q is 1.

Number of electrons in the outermost shell of R is 7.

Number of electrons in the outermost shell of S is 3.

$\therefore$ The order is CDAB.

Hence, the correct option is (b).

13. Since an ion has electrons in the outermost shell, it is more stable than an atom. An ion is a charged particle.

Hence, the correct option is (b).

14. The specific charge of α-particle is $\frac{1}{2}$ of the specific charge of the H^+ because the charge of alpha particle is twice the charge of hydrogen ion and the mass of α-particle is 4 times the mass of H^+.

Hence, the correct option is (a).

15. $\frac{5h}{2\pi}$ to $\frac{h}{\pi} \rightarrow 6$

$\frac{3h}{2\pi}$ to $\frac{h}{\pi} \rightarrow 1$

$\frac{3h}{\pi}$ to $\frac{h}{2\pi} \rightarrow 15$

$\frac{2h}{\pi}$ to $\frac{h}{\pi} \rightarrow 3$

A $\rightarrow$ (iii), B $\rightarrow$ (i), C $\rightarrow$ (iv), D $\rightarrow$ (ii)

Hence, the correct option is (a).

Assessment Test III

1. $Z = 35 \rightarrow 2, 8, 18, 7 \rightarrow$ valence electrons = 7

$Z = 15 \rightarrow 2, 8, 5 \rightarrow$ valence electrons = 5

$Z = 14 \rightarrow 2, 8, 4 \rightarrow$ valence electrons = 4

$Z = 31 \rightarrow 2, 8, 18, \rightarrow 3$ valence electrons = 3

$Z = 30 \rightarrow 2, 8, 18, 2 \rightarrow$ valence electrons = 2

The required order is DACEB.

Hence, the correct option is (d).

2. $R_n = 0.529 \times n^2, E_n = -\frac{13.6}{n^2}$ eV

(C) $n^2 = \frac{8.464}{0.529} = 16, n = 4$

(D) $n^2 = \frac{-13.6}{-1.51} = 9, n = 3$

(A) $0.529 \times n^2 = 2.116$

$n^2 = \frac{2.116}{0.529} = 4, n = 2$

(B) $0.529 \times n^2 = 0.53$

$n = 1$

The required order is CDAB.

Hence, the correct option is (c).

3. Radioactive elements emanate other particles also along with α-particles. Except α-particles, all other radioactive emanations can penetrate through lead block. The source of α-particles is kept in a lead block in order to retain α-particles alone required for the experiment.

Hence, the correct option is (a).

4. When an electron approaches an atom from an infinite distance, it loses energy. The more the distance travelled, the more the loss of energy is. Hence, potential energy decreases with the decrease in distance from the nucleus. Velocity of an electron decreases with increase in distance from the nucleus. But this has nothing to do with potential energy. Kinetic energy of an electron decreases with increase in the distance from the nucleus.

Hence, the correct option is (b).

5. Angular momentum of electron in 6th orbit = $\frac{6h}{2\pi}$

Angular momentum of electron in 2nd orbit = $\frac{2h}{2\pi}$

Difference in angular momentum during electron transition = $\frac{6h}{2\pi} - \frac{2h}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi}$.

Hence, the correct option is (b).

6. X^{-2} ion $\rightarrow$ Mass number = 32; protons = 16, neutrons = 16. Since number of protons is 16, number of electrons in neutral atom is also 16. X^{-2} ion should have 18 electrons.

   Hence, the correct option is (a).

7. (i) $\rightarrow$ (E); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A); (v) $\rightarrow$ (C)

   Hence, the correct option is (c).

8. $X \rightarrow 2, 8, 18, 8, Zx = 36.$

   Hence, the correct option is (c).

9. Number of protons = $\frac{36740}{1837} = 20$

   Atomic number = 20, Electronic configuration is 2, 8, 8, 2.

   It is calcium. It can form dispositive ion with electronic configuration of 2, 8, 8. It possesses only 8 electrons in the outermost shell.

   Hence, the correct option is (c).

10. Energy of electron in Bohr's' orbit can be given as

    $E = \frac{-13.6}{n^2} \times Z^2$, 'H' atom $Z = 1$, $E = \frac{-13.6}{n^2} = \times$ Joules

    He^+ ion $Z = 2$, $E = \frac{13.6}{n^2}$ $\therefore Z^2 = x \times 2^2 = 4x$

    Hence, the correct option is (c).

11. Charges on oil drops A, B, and C are 1.2C, 2.8C, and 0.72C, respectively.

    HCF = 0.08

    Number of electrons on A = $\frac{1.2}{0.08} = 15$

    Number of electrons on B = $\frac{2.8}{0.08} = 35$

    Number of electrons on C = $\frac{0.72}{0.08} = 9.$

    Hence, the correct option is (b).

12. Average atomic mass =

    $$\frac{(16 \times 81) + (17 \times 9) + (18 \times 10)}{100}$$

    = 16.29 amu

    Hence, the correct option is (c).

13. α-particle has 2 protons, 2 neutrons.

    Charge on α-particle = $2 \times 32 \times 10^{-19}$ C

    Mass of α-particle = $(2 \times 8.35 \times 10^{-28}$ kg$) + (2 \times 8.6 \times 10^{-28})$ kg

    Specific charge (e/m) =

    $$\frac{2 \times 3.2 \times 10^{-19}}{(2 \times 8.35 \times 10^{-28}) + (2 \times 8.6 \times 10^{-28})}$$

    $$= \frac{6.4 \times 10^{-19}}{33.9 \times 10^{-28}} = 1.9 \times 10^{8} \text{ C/kg}$$

    Hence, the correct option is (c).

14. ${}_{92}X^{234} \rightarrow {}_{90}Y^{230} + {}_{2}He^{4}$

    Hence, the correct option is (d).

15. High voltage is essential to cause ionization in the discharge tube. Low pressure should be maintained for the gas inside the discharge tube so that intermolecular forces of attraction are less and ionization of gas takes place appreciably.

    Hence, the correct option is (b).

Assessment Test IV

1. $Z = 16$; 2, 8, 6, M shell $\rightarrow$ 6 electrons

   $Z = 22$; 2, 8, 10, 2, M shell $\rightarrow$ 10 electrons

   $Z = 26$; 2, 8, 14, 2, M shell $\rightarrow$ 14 electrons

   $Z = 28$; 2, 8, 16, 2, M shell $\rightarrow$ 16 electrons

   $Z = 33$;2, 8, 18, 2, M shell $\rightarrow$ 18 electrons

   The required order is CBADE.

   Hence, the correct option is (a).

2. (D) $n = 2$, $Be^{+3} \rightarrow E = -\frac{13.6}{2^2} \rightarrow 4^2 = -54.4$ ev

   (C) $n = 2$, $Li^{+2} \rightarrow E = -\frac{13.6}{2^2} \rightarrow 3^2 = -30.6$ ev

   (B) $n = 2$, $He^{+} \rightarrow E = -\frac{13.6}{2^2} \rightarrow 2^2 = -13.6$ ev

   (A) $n = 2$, $H \rightarrow E = -\frac{13.6}{2^2} \rightarrow 1 = -3.4$ ev

   The required order is DCBA.

   Hence, the correct option is (b).

3. All the α-particles are held in the lead block as they cannot penetrate through lead block. They are allowed to eject out through the perforation on one side of the block and hit the gold foil. α-particles produce scintillations on the ZnS screen due to the fluorescent nature of ZnS.

   Hence, the correct option is (b).